EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

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NAAC GRADE A

Department of Computer Science & Engineering

CERTIFICATE

Advanced UI/UX | V Semester B.E.

This is to certify that

Mohammad Kaif

bearing USN 1EP24CS083, Section B, has successfully completed the Advanced UI/UX activity (Problem No. 34) titled

"Smart City Grievance Reporting"

has submitted this report as required on 11-11-2026, during the academic year 2026-27.

_____

Prasanth
Assistant Professor,
Dept. of CSE
Date of Submission:

Abstract

This project presents CityVoice, a smart city grievance reporting platform designed to help residents report everyday civic problems such as potholes, uncollected garbage, broken streetlights, water leakage and blocked drains, and to follow what happens to each complaint until it is resolved. The intended users are urban residents who want a quick and trustworthy way to raise a problem, along with the municipal staff who receive and act on those complaints. The design responds to the risk that conventional grievance portals use long forms, unclear categories and vague status messages, so citizens do not know whether a complaint was received, who is handling it or when it was closed. CityVoice uses a photo-first reporting flow, automatic location capture with an adjustable map pin, category suggestions, a ticket ID for every complaint, a plain-language status timeline, nearby-issue visibility, citizen confirmation of resolution and learner-style control of privacy and notification settings. The work was carried out as a design-stage case study: user journeys, low-fidelity wireframes, a visual design system and high-fidelity screen mockups were prepared for the main screens. The design is not connected to authentication, a database, a live map service or any municipal back-end system, and it has not been developed as a coded application. Formal interviews and usability tests with target users remain to be completed; therefore, no measured user-testing outcome is claimed in this report.

Project at a glance

Item	Description
Project title	Smart City Grievance Reporting
Product name	CityVoice
Domain	Smart Cities and e-Governance (Civic Technology)
Platform / prototype	Mobile-first responsive interface design (wireframes and high-fidelity screen mockups)
Technologies	Design-stage deliverables only; no coded implementation
Main design principle	Make reporting simple and make resolution visible and accountable

1. Problem Definition and Scope

Stage: Discover — define the design challenge and boundaries.

Problem statement

Residents of fast-growing cities often find it difficult to report civic problems. Many grievance systems require long forms, ask for details the citizen may not know (such as the responsible department or ward number), and give little feedback after submission. Citizens may then repeat the same complaint, lose trust in the process or stop reporting altogether. The project proposes a reporting experience that makes a complaint quick to submit, gives it a visible ticket and status, and lets the citizen confirm whether the problem was actually fixed.

Stakeholders and target users

- Primary users:residents ofacityor town who use a smartphone or web browser to report and track civic problems.
- Secondary stakeholders: municipal ward officers, field staff and department supervisors who receive, assign and resolve complaints.
- Other stakeholders: city administrators, senior citizens and people with disabilities who need accessible reporting, and people responsible for privacy and data protection.

Goals and success metrics

•Help a citizen report an issue in a few clear steps using a photo, an automatic location and a category.
•Give every complaint a ticket ID and a status timeline written in plain language.
•Show which team is responsible and let the citizen confirm or reopen a resolved complaint.
•Allow citizens to control notifications, language and whether their name is visible publicly.

Proposed metrics for later testing: task completion rate, time to submit a complaint, form errors, ease-of-use rating, and whether users understand the meaning of each status stage. These are evaluation targets, not measured results.

Constraints and assumptions

- The project is adesign-stage case study with sample data; no login, cloud database, live map service, real complaint records or municipal workflow is connected. The interface assumes a citizen has a smartphone with a camera and may have an unstable internet connection.

- The sample ticket numbers, wards, department names and complaint titles shown in the screens are demonstration content.

Scope: what the design will and will not cover

Included: home screen, report-an-issue flow (photo, location, category, description, review and submit), my complaints list, complaint status timeline, nearby issues map, notifications, confirm or reopen action, settings and privacy preferences, help and responsive navigation. **Not included:** production authentication, server-side data, integration with municipal departments, field-officer or administrator applications, payments, analytics infrastructure or actual alert delivery.

2. User Research

Stage: Discover — primary research is required before treating assumptions as validated findings.

Research method and questions asked

Recommended method: conduct short semi-structured interviews or observe at least five residents from the intended target group. Ask the same core questions and record answers only after obtaining consent.

- How do you currently report or complain about a civic problem such as a pothole or garbage?
- What stops you from reporting a problem, or makes you give up halfway?
- Have you ever reported a problem and never heard back? How did that feel?
- What information would you expect to see after your complaint is submitted?
- How comfortable are you sharing your photo, location and personal details?
- What language, notification and privacy controls would you expect?

Participant details

No interview records or survey responses were available while preparing this report. Complete the table below after speaking with five real participants. Do not enter invented names or responses.

No.	**Participant profile**	**Method**	**Key observation** [Record
1	[Age range / occupation; no full name]	[Interview / observation]	actual response] [Actual
2	[Profile]	[Method]	observation] [Actual
3	[Profile]	[Method]	observation] [Actual
4	[Profile]	[Method]	observation] [Actual
5	[Profile]	[Method]	observation]

Competitor or existing solution analysis

A formal competitor review has not been documented for this project. For the next research step, compare two or three existing civic grievance or complaint-reporting products using the same criteria: ease of submission, photo and location support, clarity of status tracking, transparency about the responsible team, language support, accessibility and privacy communication. Record product names and evidence after reviewing them.

Insights that will drive the design

- Design direction already expressed in the mockups: reporting should need as few typed fields as possible, with the photo and location doing most of the work.
- Every complaint should receive a ticket ID and a visible status so that citizens never have to guess what happened.
- A complaint should be closed only after the citizen has had the chance to confirm or reopen it.
- Privacy, language and notification controls should be visible and easy to change.

3. Personas and Empathy Map

Stage:Define—provisionalpersonastovalidate with Section 2 research.

Persona 1 — The busy commuter (provisional)

Profile: a working adult who notices problems such as potholes or broken streetlights during daily travel. Goal: report the problem within a minute without filling a long form. Possible frustration: not knowing whether anyone received the complaint. Needs: camera-based reporting, automatic location, a ticket ID and short status updates. These are design hypotheses, not findings from completed interviews.

Persona 2 — The first-time digital user (provisional)

Profile: a resident, possibly a senior citizen, who is less confident with apps and prefers a regional language. Goal: report a local problem and understand what happens next. Possible frustration: technical terms, small text and unclear buttons. Needs: simple language, large touch targets, language choice and clear confirmation messages. Validate these attributes through real participant research.

Empathy map — working hypotheses

Says	Thinks	Does	Feels
"I reported this last month and nothing happened."	"Will anyone see my complaint? Who is responsible?"	Takes a photo, searches for the right category, checks again later or complains elsewhere.	May feel ignored when there is no update and relieved when progress is visible.

Pain points and unmet needs

- Long forms and unclear categories that make reporting slow.
- No ticket ID, status or responsible team after submission.
- Complaints marked closed without the citizen knowing whether the problem was fixed.
- Concern about sharing name, phone number and exact location publicly.

Problem reframed as a design question

How might we help residents report civic problems quickly and then trust the process through clear status updates, accountability and privacy control, without making reporting feel slow, confusing or pointless?

4. User Journey and Information Architecture

Stage: Define — map the proposed journey.

End-to-end user journey with emotional highs and lows

Step	User action	Interface support	Possible feeling
1. Notice	Sees a civic problem	Quick access to Report an issue on the home screen	Frustrated
2. Capture	Takes a photo and confirms location	Camera prompt, auto-detected location and adjustable pin	Hopeful / focused
3. Describe	Selects a category, adds optional text	Category chips and optional description	In control
4. Submit	Reviews and submits	Summary screen, ticket ID and confirmation	Reassured
5. Track	Opens My complaints	Status timeline and responsible team	Informed
6. Wait	Waits for action	Alerts on status changes, user-controlled	Patient / anxious if no update
7. Close	Confirms or reopens the complaint	Confirm resolved / Reopen actions and feedback	Satisfied or heard

Task flows for the three most critical tasks

- **Report an issue:** Home → Report an issue → Add photo → Confirm location → Choose category → Review and submit → Ticket ID.
- **Track a complaint:** Home or My complaints → Select complaint → View status timeline and responsible team.
- **Confirm or reopen:** Notification → Complaint details → Confirm resolved OR Reopen with a reason.

Sitemap or navigation structure

Mainnavigation	Keycontent / actions
Home	Report an issue button, category shortcuts, recent complaints, help
Report an issue	Photo, location with map pin, category, description, privacy toggle, review and submit
My complaints	All, open and resolved filters, search by ticket ID, complaint cards
Complaint details	Ticket ID, status timeline, responsible team, add comment, confirm or reopen
Nearby issues	Map and list of public complaints near the user, category and status filters
Notifications	Status updates and confirmation requests
Profile and settings	Edit profile, language, alert preferences, quiet hours, name and location visibility

Content and feature prioritisation

Priority 1: report-an-issue flow,ticketID, complaint tracking and status timeline. Priority 2: confirm or reopen, notifications and privacy settings. Priority 3: nearby issues map, comments and help content. All these areas are covered at design level in the mockups.

5. Ideation and Concept Selection

Stage: Ideate — concepts are design directions; no scanned sketches are included.

Rough sketches or rapid ideas

This reportcontains digitally prepared wireframes and mockups rather than paper sketches or scanned ideation pages. Add photographs or scans of original sketches here if they were created during the design process.

At least three distinct concepts

Concept	Description	Strength	Risk
A — Traditional complaint form	Long web form with department, ward, category and address fields	Collects detailed information	Slow, error-prone and discouraging for ordinary citizens
B — Photo-first reporting with transparent tracking	Photo, auto location and category in a few steps, ticket ID and status timeline	Fast, accountable and easy to understand	Depends on camera and location access; needs a fallback
C — Chat-based assistant	Citizen reports through a conversational assistant	Friendly and flexible	Harder to scan, edit and track; may misread the problem

Evaluation of concepts against criteria

Criteria	Concept A	Concept B	Concept C
Speed and ease of reporting	Low	High	Medium
Status transparency	Low	High	Medium
Accessibility for varied users	Medium	High	Medium
Accountability and closure	Medium	High	Low
Overall fit with problem	Low	High	Medium

Chosen concept and reason for selection

Concept B was selected and isrepresented by CityVoice. It aligns with the central requirement: let residents report civic problems quickly and trust what happens next. The design uses a photo-first report flow, automatic location, a ticket ID, a plain-language status timeline, citizen confirmation of closure and user-controlled privacy settings.

6. Low-Fidelity Wireframes

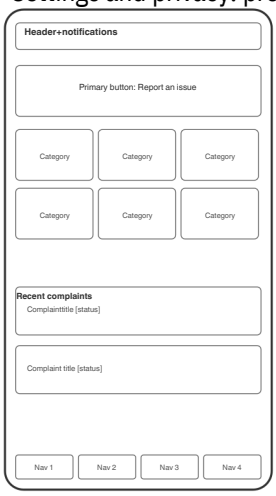
Stage: Prototype 1 — document the layout before visual styling.

Wireframes of key screens

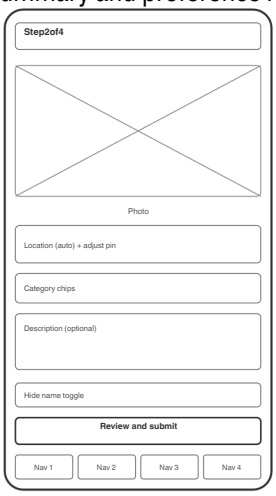
The monochrome wireframes below show the layout of four key screens before colour and branding were applied. The full screen inventory is:

- Home: header, notification icon, primary Report button, category shortcuts and recent complaints.
- Report an issue: step indicator, photo area, location with map pin, category chips, optional description, privacy toggle and submit button.
- Complaint status: ticket summary, responsible team, four-stage timeline, confirm and reopen actions.

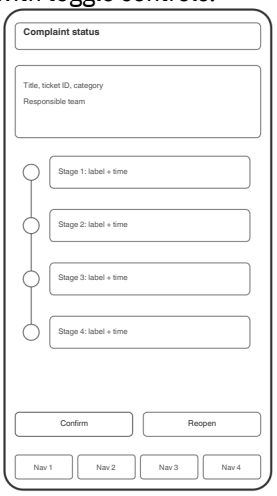
- Nearby issues: map with pins, filters and a list of nearby public complaints.
- Settings and privacy: profile summary and preference rows with toggle controls.



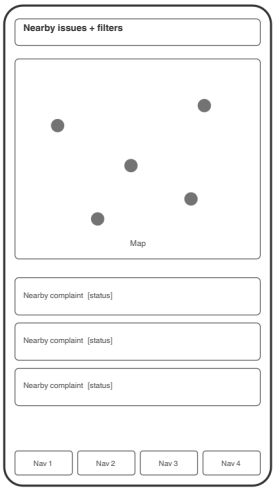
Home



Report an issue



Complaint status



Nearby issues

Annotations explaining layout and interaction decisions

- A persistent bottom navigation keeps the four main areas within thumb reach on phones; the same links move to a sidebar on larger screens.
- The Report button is the most prominent element on the home screen because reporting is the most frequent task.
- Actions sit next to the content they affect: confirm and reopen appear inside the complaint, not in a separate menu.
- The status timeline is vertical and uses text labels as well as icons so that status is never shown by colour alone.

Quick feedback from at least two people

No feedback records were available. Obtain feedback from at least two target users after showing the wireframes. Record what each person understood, where they hesitated and what they suggested.

Changes made after feedback

Pending real feedback. Do not claim that a change was user-validated until the feedback and resulting revision are recorded.

7. Visual Design and Design System

Stage: Prototype 2 – visual choices defined for the interface.

Colour palette with contrast ratios

Role	Colour	Use in interface	Contrast ratio (calculated)
Primary blue	#1D4ED8	Primary actions, active navigation, links	White text on blue: 6.7:1
Deep navy	#12263F	Main headings and body text	On white: 15.3:1
Soft blue	#EAF1FB	Highlighted areas, icon backgrounds	Navy text on soft blue: 13.4:1
White	#FFFFFF	Cards and content surfaces	Reference surface
Resolved green	#0F7B5F	Resolved / positive status text	On white: 5.2:1
In-progress amber	#B45309	Assigned / in-progress status text	On white: 5.0:1
Alert red	#B42318	Errors, location pin, reopen warnings	On white: 6.6:1
Light neutral	#F5F7FA	Pagebackground	Navytext onneutral:14.2:1

The ratios above were calculated from the hex values using the WCAG relative-luminance formula. WCAG 2.2 AA requires 4.5:1 for normal text and 3:1 for large text. These calculations should still be re-checked in the final rendered screens with a contrast checker before claiming compliance.

Typography and type scale

Proposed typeface: a clean, highly legible sans-serif (for example Inter or Noto Sans, with the system sans-serif as fallback), chosen for readability on small screens and support for Indian-language companions. Proposed scale: 12 px captions, 14 px secondary text, 16 px body, 20 px section titles and 24 px screen titles, with a minimum of 16 px for primary reading text.

Spacing, grid and layout rules

The designuses an 8-point spacing system, a single-column mobile layout with 16 px side margins, card-based content panels, rounded corners and a fixed bottom navigation. Interactive elements are intended to be at least 44 × 44 px so they are easy to tap. Exact tokens should be extracted into the design tool before final hand-in.

Reusable components and their states

- Navigationitems: active and inactive.
- Buttons: primary, secondary, text and disabled (for example, Confirm resolved before the status is Resolved).
- Cards: category shortcut, complaint card, status timeline, map list item and settings row.
- Status chips: Submitted, Assigned, In progress, Resolved and Reopened, each with a text label.
- Controls: search field, filter chips, photo picker, map pin, toggles; feedback via toast messages and a help modal.

Iconography and imagery choices

Simple outlinedicons arepaired withtextlabels for categories and status so that meaning does not depend on the icon alone. Photography is limited to the citizen's own complaint photo; no stock imagery is used, which keeps screens light and focused.

Rationale linking visual choices to user needs

Blue andwhite create a calm,trustworthyciviclook; green, amber and red are reserved for status and always accompanied by text; large cards and buttons support users with limited digital confidence. These are design intentions and should be checked with users and accessibility testing.

8. High-Fidelity Interactive Prototype

Stage: Prototype 3 — describe the designed screens and interactions.

Final screens for all critical flows

Screen	Designed content
Home	Greeting, Report an issue button, six category shortcuts, recent complaints with status chips
Report an issue	Step indicator, photo area, auto-detected location with adjustable pin, category chips, optional description, hide-name toggle, review and submit
Complaint status	Ticket ID, category, responsible team, four-stage timeline, confirm resolved and reopen actions
Settings and privacy	Profile card, five preference toggles (alerts, weekly summary, name visibility, location sharing, quiet hours) and save button

Two further screens, My complaints and Nearby issues, follow the same components and are shown at wireframe level in Section 6.

Interactions, transitions and micro-interactions

- Bottom navigation switchesbetweenthe main sections and highlights the active item.
- Tapping the photo area opens the camera or gallery; a thumbnail replaces the placeholder once a photo is added.
- Location is detected automatically, and the citizen can drag the pin or type an address to correct it.
- Submitting shows a confirmation screen with the ticket ID and a link to track the complaint.
- A status change triggers an alert according to the user's preferences; Confirm resolved is enabled only at the Resolved stage.
- Saving preferences shows a short confirmation message; the help icon opens a modal.

These interactions are specified in the design. They have not been coded and need to be demonstrated in a clickable prototype.

Edge states: empty, loading, error, offline and permission denied

- Empty:MycomplaintsshowsafriendlymessageandaReportbuttonwhenthere are no complaints.

- Loading: skeleton cards while complaints load; a progress indicator while a photo uploads.
- Error: a plain-language message with a Retry button; the typed description is kept.
- Offline: the report is saved as a draft and sent when the connection returns.
- Permission denied: if location access is refused, the citizen can search or type an address manually.

Prototype link and access instructions

No hosted or clickable prototype URL is included. The designed screens are shown on the final page of this report. To extend this work, build the screens as a clickable prototype in a design tool such as Figma, link the screens using the flows in Section 4 and add the share link here.

Tools and resources used

Wireframes and screen mockups prepared for this report; WCAG 2.2 guidelines for accessibility targets. A design tool such as Figma is recommended for the interactive prototype.

9. Usability Testing and Iteration

Stage: Test — testing is a required next step; no test results are available.

Test plan: tasks, participants and success criteria

Recruit at least five residents from the target group. Ask each participant to complete the same tasks without coaching. Obtain consent before recording observations.

- Report a pothole using a photo, location and category from the home screen.
- Find the ticket ID and current status of an existing complaint.
- Explain what the “Assigned” and “In progress” statuses mean.
- Confirm that a resolved complaint is fixed, or reopen it with a reason.
- Open settings, hide your name publicly and change one notification preference.

Success criteria: participant completes the intended task without assistance, identifies the expected result and can explain the main interaction. Record completion, time, errors and a simple satisfaction rating.

Results: task success, time on task, errors and satisfaction

Not yet measured. The table is intentionally blank so that real observations can be entered after testing.

User	Task	Success (Y/N)	Time	Issues observed
P1	[Task]	[ ]	[ ]	[Actual issue]
P2	[Task]	[ ]	[ ]	[Actual issue]
P3	[Task]	[ ]	[ ]	[Actual issue]
P4	[Task]	[ ]	[ ]	[Actual issue]
P5	[Task]	[ ]	[ ]	[Actual issue]

[Actual issue]

Accessibility check and issues fixed

A complete WCAG 2.2 AA audit is not documented. Review keyboard and screen-reader access, visible focus, semantic labels, colour contrast in the final screens, text resizing, touch target size, language support and reduced-motion behaviour. Fix issues and record before/after evidence.

Problems found, ranked by severity

No observed user problems can be ranked until the usability test is conducted. Potential areas to inspect include category clarity, understanding of status names, willingness to grant location access, discoverability of the Reopen action and clarity of privacy options.

Design changes made and evidence of improvement

Pending test results. Add each issue, severity, change made and retest outcome after testing.

10. Design Rationale, Reflection and Future Work

Stage: Reflect — evaluate the design honestly.

Major design decisions and trade-offs

- Photo-first reporting with automatic location: reduces typing and errors, but depends on camera and location access, so manual fallbacks are needed.
- Plain-language status timeline with a responsible team: builds trust and accountability, but depends on departments updating statuses honestly and on time.
- Citizen confirmation before closure: helps prevent false closure, but needs rules for complaints the citizen never confirms.
- Hide-name and area-only location options: protect privacy, but anonymous reports can make follow-up questions harder.
- Design-stage deliverables only: made it possible to define the full experience without back-end setup, but it does not store or process real complaints.

Limitations of the final design

- Sample content is used; nothing is connected to a real city system, complaint database or department workflow.
- Authentication, notifications, maps and persistent storage are not implemented.
- The interactions and edge states are specified but not coded or demonstrated in a clickable prototype.
- No formal participant research, usability test or full accessibility audit is included.
- Multi-language support is a design intention; translated content has not been prepared or checked.

What failed or was redone, and what was learned

No design-process log describing failed concepts or redesign iterations was kept. Based on the work done, the main learning is that a clear reporting flow can be defined and visualised before development, while real behaviour such as routing, department response and notifications needs integration with municipal systems and user testing.

Ethical considerations

- Collect only the data needed to resolve a complaint; explain why photo, location and contact details are requested and obtain consent.
- Warn users not to capture faces or vehicle number plates, and allow hiding of the citizen's name from public view.
- Present accountability at department level and avoid publicly blaming individual staff members.
- Design for equal access: simple language, regional-language support, large touch targets and non-smartphone alternatives.
- Plan moderation for duplicate, false or abusive reports without discouraging genuine ones.

Future improvements

- Conduct five or more user interviews and usability tests, then revise the personas and flows using actual evidence.
- Build a clickable prototype and, later, connect authentication, secure storage and map services.
- Integrate with department workflows so that complaints are routed and status updates are real.
- Add explicit loading, error, offline and permission-denied states and test them.
- Complete WCAG 2.2 AA checks and test with keyboard and assistive technology users.
- Validate that notifications and status messages remain helpful rather than overwhelming.

References

[1] CityVoicedesign materials (wireframes, mockups and design specification) prepared for this report, accessed 11 Nov. 2026.

[2] UI/UX Project Report format, Advanced UI/UX | V Semester B.E., supplied course reference report.

[3] World Wide Web Consortium (W3C), "Web Content Accessibility Guidelines (WCAG) 2.2," W3C Recommendation, Oct. 2023. [Online]. Available: https://www.w3.org/TR/WCAG22/

Appendix

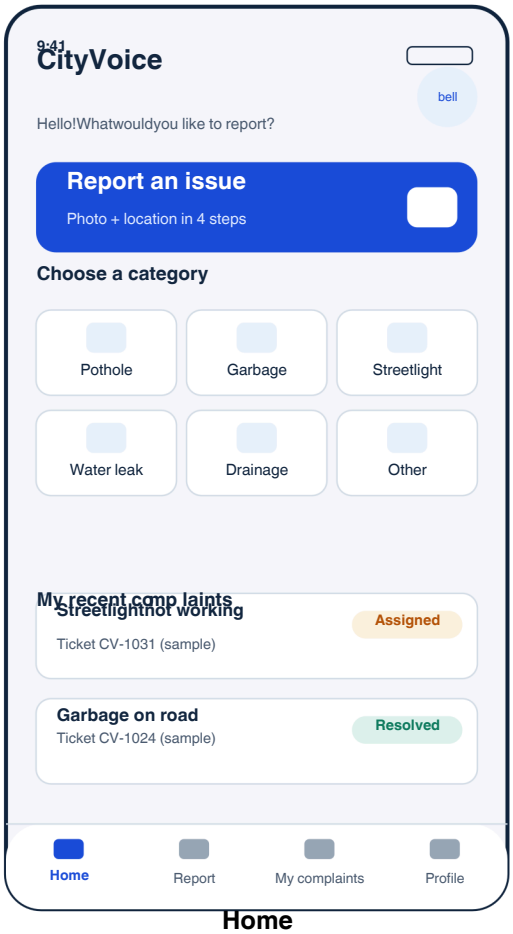
- Interview scripts and consent notes — to be added after conducting participant research.
- Raw research notes and survey responses — to be added; no participant data was available for this report.
- Usability test scripts and completed task records — to be added after testing.
- Additional screens and sketches — insert any original low-fidelity sketches and the My complaints and Nearby issues screens.

Submission checklist

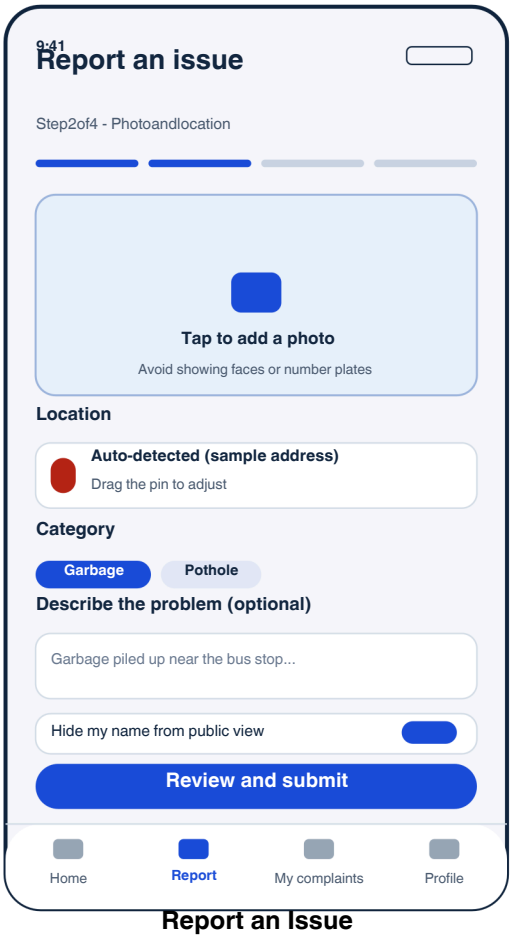
Item	Action before final submission
Student details	Certificate filled with name, USN, section, problem number and submission date; confirm guide name and signature.
Research evidence	Interview at least five target users and complete Section 2.
Personas	Update provisional personas using real findings.
Wireframes	Add any hand sketches and the feedback collected on the wireframes.
Accessibility	Re-check contrast on final screens and document WCAG 2.2 AA checks.
Usability testing	Test with at least five users and complete the results table.
Prototype evidence	Add a clickable prototype link and screenshots if available.

CityVoice – Prototype Output Screens

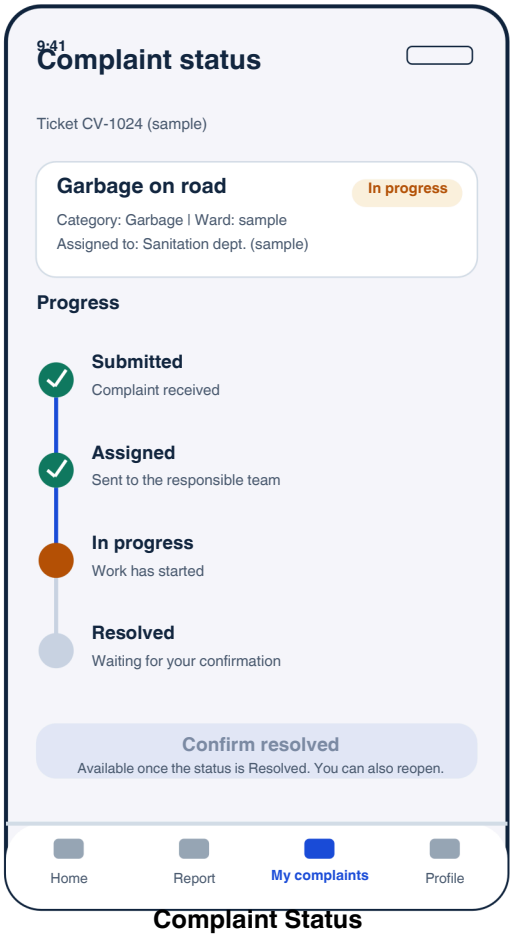
Designed screens ofthepropose dinterface(high-fidelitymockups,notscreenshotsofacodedapplication; all data is sample content)



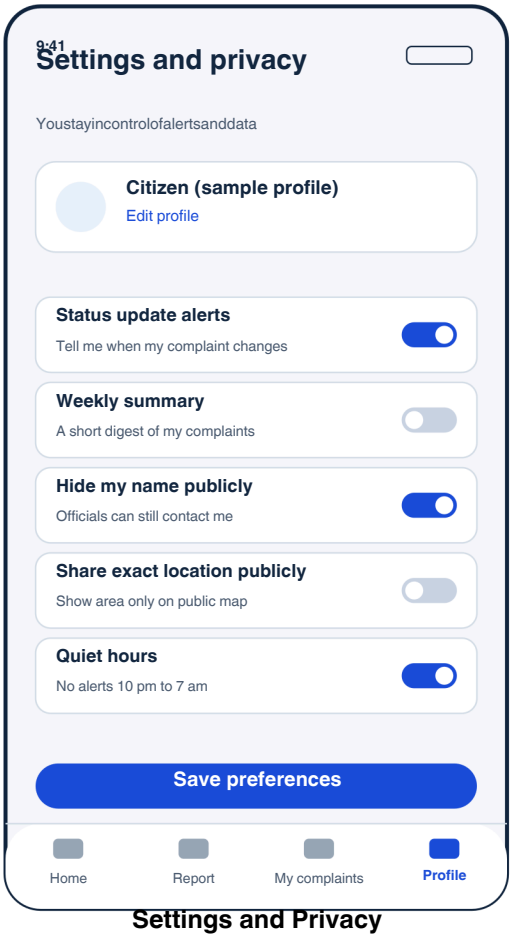
Home



Report an Issue



Complaint Status



Settings and Privacy